

AI Engineer Data Intelligence Pipeline — System Architecture

Executive Overview: The AI Engineer Data Intelligence Pipeline is a resilient, fault-tolerant, and traceable data extraction system designed to ingest AI domain knowledge (Startups, Products, Research Papers, News, and Jobs) with zero data hallucination and strict provenance verification.

1. System Architecture Breakdown

Module	Components	Key Responsibilities & Guarantees
Crawler (src/crawler/)	http_client.py playwright_client.py github_client.py crawler.py	Asynchronous HTTP (<code>`aiohttp`</code>) + Playwright fallback for JS pages. Bounded concurrency (<code>`Semaphore`</code>), per-domain delays, exponential backoff with full jitter on HTTP 429, GitHub REST API star verification.
Parsers (src/parsers/)	date_parser.py paper_parser.py news_parser.py job_parser.py	Deterministic extraction using BeautifulSoup, OpenGraph, and JSON-LD. Converts dates to ISO-8601 UTC. Evaluates 24-hour freshness rule for news and job postings.
LLM Engine (src/llm/)	base_provider.py gemini_provider.py groq_provider.py deepseek_provider.py orchestrator.py chunker.py	Unified provider contract with automated fallback chain: Gemini (Primary) → Groq (Fallback 1) → DeepSeek (Fallback 2) . Enforces JSON output. Content chunking on HTTP 413.
Resolver (src/resolver/)	entity_resolver.py	Normalizes company names (strips <i>Inc, LLC, Corp, Ltd</i>), performs alias lookups, sequence matcher fuzzy scoring, and logs resolution confidence.
Storage & Export (src/storage/ src/exporters/)	database.py models.py repository.py validators.py csv_exporter.py	SQLAlchemy ORM models on SQLite (<code>`data/app.db`</code>). Enforces URL deduplication and schema validation (pricing enums). Generates pandas CSV exports in <code>`data/exports/`</code> .

2. LLM Provider Fallback Chain & Resilience

When semantic extraction is required, the system executes an automated fallback chain across configured providers without terminating extraction:

[Gemini (google-genai)] —(failure / no key)→ **[Groq (AsyncGroq)]** —(failure / no key)→ **[DeepSeek (OpenAI API)]**

3. Production Scalability & Anti-Bot Strategy

- **Data Provenance:** Every record preserves source name, source URL, and collection timestamp (``collectedAt``). Missing values default strictly to null.
- **Anti-Bot Compliance:** Respects robots/terms, uses official REST APIs (GitHub API), avoids illegal CAPTCHA bypasses, and applies jittered exponential backoff.
- **PostgreSQL Migration:** Change ``DATABASE_URL`` in ``.env`` to ``postgresql://...`` to scale to production without modifying source code.